Kuan-Wen Lai

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Empolyment & Positions

National Center for Theoretical Science, Taiwan

2025.03–present, affiliated member

Tunghai University, Taiwan

2024.08.01–present, Assistant Professor

Academia Sinica, Taiwan

2023.09.01–2024.07.31, Postdoc

Universität Bonn, Deutschland

2021.09.01–2023.08.31, Postdoc

Part of the ERC Synergy Grant HyperK (10.3030/854361).

University of Massachusetts Amherst, USA

2018.09.01–2021.08.31, Visiting Assistant Professor

R.O.C. (Taiwan) Army

2011.08.08–2012.07.09, obligatory military service

Education

Brown University, USA

Ph.D. Mathematics, 2018.05.27

Thesis: *Cremona transformations and rational parametrizations inspired by Hodge theory*.

Advisor: Brendan Hassett

Rice University, USA

2013.08–2015.07, graduate student

Transferred to Brown University with advisor Brendan Hassett in the summer of 2015.

Taida Institute for Mathematical Sciences, Taiwan

2012.08–2013.07, research trainee

National Taiwan University, Taiwan

M.S. Mathematics, 2011.06.04

Thesis: *The Gromov-Witten Theory of* $\mathbb{P}^1$.

Advisor: Hui-Wen Lin

B.S. Physics, with minor in Mathematics, 2009.06.07

Grants

- Academic Summit Program 114-2639-M-002-009-ASP (CI), 2025.08.01–2030.07.31
- NSTC Research Project 113-2115-M-029-003-MY3 (PI), 2024.10.01–2027.07.31

Research

Algebraic Geometry, with a focus on Fano fourfolds with K3-type Hodge structures, and on Cremona transformations from derived-categorical perspective. I have also worked on algebraic geometry over finite fields and its applications to coding theory.

Preprints

- A geometric approach to the density of rank-metric codes (with Shamil Asgarli, Lian Duan, and Nathan Kaplan), https://arxiv.org/abs/2609.21189.
- Mirror symmetry for lattice-polarized abelian surfaces (with Yu-Wei Fan), https://arxiv.org/abs/2602.04232.

Publications

1. Categorical Nielsen realization problems for generic K3 surfaces (with Yu-Wei Fan), Journal of Differential Geometry, accepted.
2. Fourier–Mukai numbers of K3 categories of very general special cubic fourfolds (with Yu-Wei Fan), Israel Journal of Mathematics 268, 427–437 (2025).
3. On the irrationality of moduli spaces of projective hyperkähler manifolds (with Daniele Agostini and Ignacio Barros), Épijournal de Géométrie Algébrique, Volume 9 (2025), Article No. 13.
4. Frobenius nonclassical hypersurfaces (with Shamil Asgarli and Lian Duan), Mathematische Zeitschrift 309, 18 (2025).
5. Transverse linear subspaces to hypersurfaces over finite fields (with Shamil Asgarli and Lian Duan), Finite Fields and Their Applications 95 (2024), Paper No. 102396.
6. New rational cubic fourfolds arising from Cremona transformations (with Yu-Wei Fan), Algebraic Geometry 10 (4) (2023), 432–460.
7. On the irrationality of moduli spaces of K3 surfaces (with Daniele Agostini and Ignacio Barros), Transactions of the American Mathematical Society 376 (2023), No. 2, 1407–1426.
8. Uniform potential density for rational points on algebraic groups and elliptic K3 surfaces (with Masahiro Nakahara), International Mathematics Research Notices 2022, no. 23, 18541–18588.
9. Decompositions of Lagrangian classes on K3 surfaces (with Yu-Shen Lin and Luca Schaffler), Mathematical Research Letters 28 (2021), no. 6, 1739–1763.
10. Bijective Cremona transformations of the plane (with Shamil Asgarli, Masahiro Nakahara, and Susanna Zimmermann), Selecta Mathematica, New Series 28, 53 (2022).
11. Transverse lines to surfaces over finite fields (with Shamil Asgarli and Lian Duan), manuscripta mathematica 165, 135–157 (2021).
12. Cremona transformations and derived equivalences of K3 surfaces (with Brendan Hassett), Compositio Mathematica 154 (2018), Issue 7, 1508–1533.

13. New cubic fourfolds with odd degree unirational parametrizations, Algebra & Number Theory 11 (2017), No. 7, 1597–1626.

Proceedings

- A tour of the rationality problem of cubic fourfolds, Proceedings of the International Congress of Chinese Mathematicians 2020.

Graduate Advisees

- Ciou (Tunghai University), 2026–
- Hsiao (National Taiwan University), 2026–

Invited Talks

2027 Taiwanese Mathematical Society (TMS) Annual Meeting, National Taiwan University, January 18–19

- Workshop on Algebraic Geometry, Xiamen University Malaysia, January 10–16

2026 Mini-Workshop: Algebraic Geometry in Xitou, July 19–21

- 2026 Summer School in Algebraic Geometry, National Center for Theoretical Sciences (NCTS), June 29–July 17
- Tour de Formosa: Algebraic Geometry along Taiwan's eastern coast, January 12–16

2024 Algebraic Geometry in Taipei, National Taiwan University, December 2–6

- Colloquium, National Sun Yat-Sen University, March 8
- Colloquium, National Kaohsiung Normal University, March 7
- Taiwanese Mathematical Society (TMS) Annual Meeting, National Chengchi University, January 22–23
- Taipei Postdoc Seminar, Academia Sinica & National Center for Theoretical Sciences (NCTS), January 3

2023 Student Geometry Seminar, National Tsing Hua University, November 16

- Algebraic Geometry in East Asia, Korea Institute for Advanced Study (KIAS), November 6–10
- NCTS Seminar in Algebraic Geometry, National Center for Theoretical Sciences (NCTS), September 22
- Algebra, geometry and number theory seminar, Universiteit Leiden, April 26
- Higher Dimensional Algebraic Geometry Workshop, National Center for Theoretical Sciences (NCTS), March 20–24
- Colloquium, National Central University, March 16
- Colloquium, National Cheng Kung University, March 9

2022 Birational Complexity of Algebraic Varieties, Simons Center for Geometry and Physics (SCGP), December 12–16

- Géométrie Algébrique et Géométrie Complexe, Centre International de Rencontres Mathématiques (CIRM), November 28–December 2

- Algebraic and Arithmetic Geometry Seminar, Universität Bielefeld, June 1
- Colloquium, National Tsing Hua University, March 21 (Online)

2021 Arithmétique et géométrie algébrique, Université de Strasbourg, December 9
- A tour of the rationality problem of cubic fourfolds, Academia Sinica, Taipei, Lecture series during July 28–August 3
- Seminar of Algebraic Geometry in East Asia, April 16 (Online)

2020 International Congress of Chinese Mathematicians (ICCM) at the University of Science and Technology of China, December 27–29 (Online)
- Mathematical Physics Seminar, Center of Mathematical Sciences and Applications (CMSA), April 20 (Online)
- Geometry, Physics, and Representation Theory Seminar, Northeastern University, January 23
- Colloquium, National Central University, Taiwan, January 9

2019 Front Range Algebraic Geometry and Number Theory (FRAGMENT) seminar, Colorado State University, December 12
- Algebra and Number Theory Seminar, Dartmouth College, September 17
- A Summer Day on Algebraic Geometry, Center for Advanced Study in Theoretical Sciences (CASTS), July 1
- NT & AG Seminar, Boston College, February 7
- Workshop on Moduli Spaces and Related Problems, Shanghai Center for Mathematical Sciences (SCMS), September 17–21
- Valley Geometry Seminar, UMass Amherst, March 2

2017 Mini-talk at Georgia Algebraic Geometry Symposium (GAGS), University of Georgia, March 3–5
- Yale Algebraic and Tropical Geometry, Yale University, February 16
- Princeton Algebraic Geometry Seminar, Princeton University, February 14
- Algebra and Algebraic Geometry Seminars, Brown University, February 10

2016 Algebra and Algebraic Geometry Seminars, Brown University, October 14

2013 Algebraic Geometry and Number Theory Seminar, Rice University, December 3

Teaching & Related Activities

Tunghai University

2026 115-1 Asobou! Combinatorial Games
- 115-1 Calculus A (I)
- 115-1 Mission Mars (II)
- 114-2 Differential Geometry
- 114-2 Calculus A (II)
- 114-2 Undergraduate Research (9 students)

- 114-2 Mission Mars (I)
    - A course in collaboration with a science fair, as part of the ”Mission Mars” event at the National Museum of Natural Science in November each year.

2025 114-1 Asobou! Combinatorial Games

- 114-1 Calculus A (I)
- 114-1 Mission Mars (II)
- 113-2 Calculus A (II) (Two sections)
- 113-2 Undergraduate Research (6 students)
- 113-2 Mission Mars (I)
- UTAR × Tunghai International-National Mathematics Competition
    - A math competition in July 2025 co-organized by my department and the UTAR Mathematics Society of Universiti Tunku Abdul Rahman (UTAR).

2024 113-1 Calculus A (I) (Two sections)

- 113-1 Mission Mars (II)

Universität Bonn

2022 Summer Semester, Birational Geometry of Algebraic Varieties

UMass Amherst

2021 Spring, Fundamental Concepts of Mathematics (Two sections)

2020 Fall, Independent Study on Algebraic Geometry
    - Advisee: Andrew Tawfeek (Amherst College)
- Fall, Fundamental Concepts of Mathematics (Two sections)
- Spring, Independent Study on Algebraic Geometry
    - Advisee: Andrew Tawfeek (Amherst College)
- Spring, Fundamental Concepts of Mathematics
- Spring, Multivariate Calculus

2019 Fall, Multivariate Calculus (Two sections)

- Spring, Calculus II (Two sections)

2018 Fall, Calculus I (Two sections)

Brown University

2017 Fall, Intermediate Calculus

2015 Fall, Introductory Calculus I

Rice University

2015 Summer, Single Variable Calculus II